

Implicit differentiation, what's going on here?

When y cannot be separated from x .

THE IDEA

Sometimes y is tangled up with x and cannot be written as $y = \text{something}$. A circle, $x^2 + y^2 = 25$, is the classic case. The trick: differentiate **both sides** with respect to x , and remember that y is secretly a function of x — so every y picks up a **dy/dx** by the chain rule.

Term	Its derivative with respect to x
x^2	$2x$
y^2	$2y \cdot dy/dx$ — the chain rule adds the dy/dx
a constant	0
$x \cdot y$	$y + x \cdot dy/dx$ — needs the product rule too

WORKED EXAMPLE

CHAPTER 6 — WORKED EXAMPLE

implicit differentiation

THE IDEA

when y is tangled with x and cannot be separated,

differentiate BOTH sides — and every y gets a dy/dx attached

EXAMPLE $x^2 + y^2 = 25$ (a circle)

differentiate each term with respect to x :

$$x^2 \rightarrow 2x$$

$$y^2 \rightarrow 2y \cdot dy/dx$$

$$25 \rightarrow 0$$

$$2x + 2y \cdot dy/dx = 0$$

$$2y \cdot dy/dx = -2x$$

$$dy/dx = -x / y$$

WHY y GETS dy/dx

because y is itself a function of x — this is the chain rule again.

THE METHOD IN THREE STEPS

1. Differentiate every term on both sides. 2. Attach dy/dx to every term that came from a y . 3. Collect the dy/dx terms and solve for dy/dx .

Why it matters for ML: it is the same insight backpropagation relies on — variables that depend on other variables carry their derivatives with them through the chain rule.

FORMULAS FROM THIS CHAPTER

Method	differentiate both sides with respect to x
Every y	picks up a dy/dx
$d/dx (y^2)$	$2y \cdot dy/dx$
Then	collect dy/dx terms and solve

TEST — Chapter 6

- Find dy/dx for $x^2 + y^2 = 100$. $\rightarrow -x/y$
- Find dy/dx for $y^2 = 4x$. $\rightarrow 2y \cdot dy/dx = 4 \rightarrow dy/dx = 2/y$
- Find dy/dx for $x^2 + 3y = 12$. $\rightarrow 2x + 3 \cdot dy/dx = 0 \rightarrow dy/dx = -2x/3$